

PH 724 Data Analysis & Scientific Communication Incorporating Use of Generative AI

Part 3: Modeling Correlated Outcomes

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Correlated Data

Many datasets contain observations that are **not independent**.

Examples:

- Repeated measurements from the same individual
- Measurements from multiple sites on the same experimental unit
- Patients clustered within hospitals
- Students clustered within schools

Two common approaches for analyzing correlated data are:

1. **Mixed-effects models**
2. **Generalized estimating equations (GEE)**

Mixed-effects Models

Mixed-effects models include both:

- **Fixed effects:** effects of primary scientific interest
- **Random effects:** accounting for variation and correlation among observations within the same cluster

Subtypes of mixed-effects models for different outcomes:

Outcome	Possible Model
Normal	Linear mixed-effects model (LMM)
Binary	Logistic generalized linear mixed-effects model (GLMM)
Count	Poisson GLMM
Rate	Poisson/negative binomial GLMM

A simple example of LMM:

$$Y_{ij} = \beta_0 + \beta_1 X_{ij} + b_i + \epsilon_{ij}$$

where

- Y_{ij} : the j th outcome for the i th subject
- β_0, β_1 : fixed effects

- b_i : subject-specific random effect
- ϵ_{ij} : residual error

LMM vs GLMM

- LMM: used when the outcome is approximately **continuous and normally distributed**
 - Typically estimated using the maximum likelihood (ML) or restricted maximum likelihood (REML)
 - * ML: estimating fixed effects and variance components by maximizing the likelihood; useful when:
 - 1) comparing models with different fixed effects and 2) likelihood-based model comparison is needed
 - * REML: estimating variance components while accounting for the estimation of fixed effects; often preferred for:
 - 1) estimating variance components and 2) reporting the final LMM when the fixed-effects structure has been selected
- GLMM: extending LMM to **non-normal outcomes**
 - Typically estimated using using ML

GEE

For subject/cluster i , denote outcomes by

$$\mathbf{Y}_i = [Y_{i1}, \dots, Y_{im_i}]^\top$$

and let

$$\boldsymbol{\mu}_i = E(\mathbf{Y}_i) = [\mu_{i1}, \dots, \mu_{im_i}]^\top$$

in which $\mu_{ij} = E(Y_{ij})$. Assuming that

$$g(\mu_{ij}) = \mathbf{x}_{ij}^\top \boldsymbol{\beta},$$

the coefficient vector $\boldsymbol{\beta}$ is estimated by solving

$$\sum_{i=1}^N D_i^\top V_i^{-1} (\mathbf{Y}_i - \boldsymbol{\mu}_i) = 0$$

for $\boldsymbol{\beta}$, where

$$D_i = \frac{\partial \boldsymbol{\mu}_i}{\partial \boldsymbol{\beta}^\top}$$

and V_i is the working covariance matrix.

Major advantages of GEE over mixed-effects models

- GEE does **not** require specifying the full joint distribution of the outcomes.
- GEE does **not** require random effects.
- GEE directly estimates **population-averaged effects**.
- If the **mean model is correctly specified**, coefficient estimates can remain consistent even when the working correlation structure is misspecified.
- Correctly specifying the correlation structure can still improve efficiency.

Implementation in R

- LMM: `lmer()` from R package `lme4`
- GLMM: `glmer()` from R package `lme4`
- GEE: `geeglm()` from R package `geepack`